



3704 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Cycle: 2, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
BRIGHTKN - EPOCH1				
	1	NIRSpec	NIRSpec Fixed Slit Spectroscopy	(1) KN
	2	MIRI SED	MIRI Imaging	(1) KN

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NORMALKN - EPOCH2				
	3	NIRCAM SED	NIRCam Imaging	(1) KN
	4	MIRI SED	MIRI Imaging	(1) KN
FAINTKN - EPOCH3				
	5	NIRCAM SED	NIRCam Imaging	(1) KN
	6	MIRI SED	MIRI Imaging	(1) KN
TEMPLATE				
	7	NIRCAM SED	NIRCam Imaging	(1) KN
	8	MIRI SED	MIRI Imaging	(1) KN

ABSTRACT

Approximately half of the elements heavier than iron are formed by rapid neutron capture (r-process) reactions, yet the astrophysical site(s) of these processes remain one of the most enduring mystery of modern astrophysics. Compact binary mergers may be a major, if not dominant, source of r-process elements in the universe. However, it remains an open question whether they could reproduce the abundance pattern observed in the solar neighborhood, including the heaviest elements such as actinides and transuranic nuclei.

We propose non-disruptive ToO observations of a nearby ($z < 0.1$) kilonova to tackle this question and constrain the formation of heavy r-process elements in the ejecta of a compact binary merger. Heavy elements affect the bolometric luminosity of a kilonova and shape its temporal evolution on timescales of weeks-months after the merger. During this phase, the bulk of the emission shifts to infrared wavelengths, where only JWST can follow its evolution. A luminous and long-lived infrared emission is the hallmark signature of robust r-process production and represents incontrovertible evidence that these mergers enriched our universe of heavy metals.

OBSERVING DESCRIPTION

We proposed a flexible observing strategy to best match the evolution of the kilonova from bright to faint. We envision 3 epochs at approximately 30, 50 and 80 days, and a fourth epoch at much later times (> 120 d) to serve as reference image.

Epoch1 combines NIRSpec spectroscopy (PRISM/CLEAR) with MIRI imaging (F560W, F770W, F1000W).

This setup is optimal for bright and red kilonovae.

Epoch2 combines NIRCam multicolor imaging (3 SW +3 LW filters) with MIRI imaging (F560W, F770W). This setup is designed to offer rich color information for kilonovae that are too faint for spectroscopy.

Epoch3 uses NIRCam wider filters (F150W2/F322W2) and a single MIRI deep exposure (F770W) to characterize the faintest stages of the kilonova. Given that the source position, brightness and distance scale is unknown a priori, our combination of spectroscopy and broadband imaging can be

easily tweaked to best match the kilonova properties at the moment of trigger.

All our observations will be non-disruptive.

Proposal 3704 - Targets - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Generic Targets	#	Name	Criteria	Description
	(1)	KN	Kilonova at $z < 0.1$	<i>Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.</i>

Proposal 3704 - Observation 1 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 1: NIRSpec Diagnostic Status: Warning Observing Template: NIRSpec Fixed Slit Spectroscopy											Thu Jul 11 21:00:37 GMT 2024
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria		Description							
	(1)	KN	Kilonova at z<0.1		Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.							
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	F140X	NRSRAPIDD6	3	1	1	171.788	146441	
Template	Slit					Subarray						
	S200A1					FULL						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	40	2	1	NONE	6	12	7177.734	146441

Proposal 3704 - Observation 1 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Special Requirements	On Hold Wait for trigger Target Of Opportunity Response Time 18 Days, Carry-Over Group Observations 1, 2, Non-interruptible
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Proposal 3704 - Observation 2 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 2: MIRI SED										Thu Jul 11 21:00:37 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria				Description				
	(1)	KN	Kilonova at z<0.1								
	Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	POINT SOURCE	POSITIVE	DEFAULT	
	2	CYCLING	1	9						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	2	F770W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	3	F1000W	FASTR1	100	1	1	Dither 2	9	9	2497.536	
Special Requirements	On Hold Wait for trigger										
	Target Of Opportunity Response Time 18 Days, Carry-Over										
	Group Observations 1, 2, Non-interruptible										

Proposal 3704 - Observation 3 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 3: NIRCAM SED Diagnostic Status: Warning Observing Template: NIRCAM Imaging										Thu Jul 11 21:00:37 GMT 2024
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria	Description							
	(1)	KN	Kilonova at $z < 0.1$	<i>Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.</i>							
Template	Module	Subarray			Target Placement						
	ALL	FULL			Module Gap						
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F090W	F277W	SHALLOW4	6	1	4	4	1245.465		
	2	F150W	F356W	SHALLOW4	6	1	4	4	1245.465		
	3	F200W	F444W	SHALLOW4	6	1	4	4	1245.465		
Special Requirements	Offset 120.0 arcsec, 35.0 arcsec On Hold Wait for trigger Target Of Opportunity Response Time 18 Days, Carry-Over Group Observations 3, 4, Non-interruptible										

Proposal 3704 - Observation 4 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 4: MIRI SED										Thu Jul 11 21:00:37 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria	Description							
	(1)	KN	Kilonova at z<0.1								
	Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	POINT SOURCE	POSITIVE	DEFAULT	
	2	CYCLING	1	9						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	2	F770W	FASTR1	100	1	1	Dither 2	9	9	2497.536	
Special Requirements	On Hold Wait for trigger Target Of Opportunity Response Time 18 Days, Carry-Over Group Observations 3, 4, Non-interruptible										

Proposal 3704 - Observation 5 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 5: NIRCAM SED Diagnostic Status: Warning Observing Template: NIRCAM Imaging										Thu Jul 11 21:00:37 GMT 2024
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria		Description						
	(1)	KN	Kilonova at $z < 0.1$		Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.						
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module Gap				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F150W2	F322W2	MEDIUM8	8	1	4	4	3349.872		
Special Requirements	Offset 120.0 arcsec, 35.0 arcsec On Hold Wait for trigger Target Of Opportunity Response Time 18 Days, Carry-Over Group Observations 5, 6, Non-interruptible										

Proposal 3704 - Observation 6 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 6: MIRI SED										Thu Jul 11 21:00:37 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria			Description					
	(1)	KN	Kilonova at z<0.1								
	Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	CYCLING	1	18		5	1			DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F770W	FASTR1	100	1	1	Dither 1	18	18	4995.072	
Special Requirements	On Hold Wait for trigger Target Of Opportunity Response Time 18 Days, Carry-Over Group Observations 5, 6, Non-interruptible										

Proposal 3704 - Observation 7 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 7: NIRCAM SED Diagnostic Status: Warning Observing Template: NIRCAM Imaging										Thu Jul 11 21:00:37 GMT 2024
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria	Description							
	(1)	KN	Kilonova at $z < 0.1$	Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.							
Template	Module	Subarray			Target Placement						
	ALL	FULL			Module Gap						
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F090W	F277W	SHALLOW4	6	1	4	4	1245.465		
	2	F150W	F356W	SHALLOW4	6	1	4	4	1245.465		
	3	F200W	F444W	SHALLOW4	6	1	4	4	1245.465		
	4	F150W2	F322W2	MEDIUM8	8	1	4	4	3349.872		
Special Requirements	Offset 120.0 arcsec, 35.0 arcsec On Hold Wait for trigger Target Of Opportunity Response Time 30 Days, Carry-Over										

Proposal 3704 - Observation 8 - Identifying the fingerprints of heavy r-process elements with the James Webb Telescope

Observation	Proposal 3704, Observation 8: MIRI SED										Thu Jul 11 21:00:37 GMT 2024
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria				Description				
	(1)	KN	Kilonova at z<0.1								
	Comments: This proposal is not limited to gravitational wave counterparts but targets any kilonova discovered within this distance range.										
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				5	1	POINT SOURCE	POSITIVE	DEFAULT	
	2	CYCLING	1	18						DEFAULT	
	3	CYCLING	19	9						DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	100	1	1	Dither 1	4	4	1110.016	
	2	F770W	FASTR1	100	1	1	Dither 2	18	18	4995.072	
	3	F1000W	FASTR1	100	1	1	Dither 3	9	9	2497.536	
Special Requirements	On Hold Wait for trigger Target Of Opportunity Response Time 30 Days, Carry-Over										